Adjective Ordering in Referential Communication: Subjectivity, Discriminatory Strength, and Incremental Pragmatics

Hening Wang, Fabian Schlotterbeck, and and Michael Franke

University of Tübingen

Abstract

Adjective ordering preferences are robust across languages, and recent work suggests that they reflect both stable lexical-semantic asymmetries and context-sensitive pressures on efficient reference resolution. The present paper develops a Rational Speech Act model that brings these traditions together by crossing two dimensions in a 2×2 comparison: speaker architecture (global vs. incremental) and semantic regime (context-fixed vs. context-updating). Under context-updating semantics, the interpretation of a gradable adjective is recomputed from the listener's running posterior during composition, making adjective order non-commutative; under context-fixed semantics, interpretations are fixed from the scene. Two referential experiments in German provide the empirical target: a slider-rating study that measures graded preference for competing adjective orders and a free-production study that measures actual utterance choice. Across both experiments, ordering preferences vary systematically with the discriminatory value of the adjectives in the current scene, with speakers tending to place the more discriminating adjective earlier in the string, while still preserving a baseline size-first tendency. Bayesian model comparison shows that incremental production is the dominant mechanism across both datasets. In the production data, context-updating semantics provides a significant further improvement, but only when paired with an incremental speaker — an interaction indicating that the two mechanisms are synergistic. Simulation results confirm that a substantial baseline ordering effect already follows from context-sensitive restrictive composition alone. We argue that adjective ordering in referential communication is best understood as a graded adaptation to efficient, incremental reference resolution: subjectivity provides a baseline bias, discriminatory strength modulates that bias in context, and incremental production with context-updating semantics best captures how speakers realize these pressures online.

Keywords: adjective ordering, Rational Speech Act, incremental pragmatics, communicative efficiency, subjectivity

Adjective Ordering in Referential Communication: Subjectivity, Discriminatory Strength, and Incremental Pragmatics

1 Introduction

When multiple adjectives modify a noun, speakers across typologically diverse languages prefer similar orderings — "big brown bag" over "brown big bag" (Scontras, 2023; Sproat & Shih, 1991). Previous research (for an overview, see Scontras, 2023) has linked these preferences to meaning-based factors such as closeness to the noun and inherentness (Martin, 1969; Sweet, 1898; Whorf, 1945), to class-based hierarchies (Cinque, 1994; Dixon, 1982; Quirk, Greenbaum, Leech, & Svartvik, 1972; Scott, 2002), and to information-theoretic measures (Dyer, Torres, Scontras, & Futrell, 2023; Futrell, Dyer, & Scontras, 2020; Hahn, Degen, Goodman, Jurafsky, & Futrell, 2018). However, no single factor has proven sufficient; even the best individual predictors leave complementary variance unexplained (Dyer et al., 2023). Furthermore, the preferences are context-sensitive in ways that fixed hierarchies do not fully capture (Fukumura, 2018; Trotzke & Wittenberg, 2019).

We focus on two lines of recent work that ground ordering preferences in communicative efficiency. Scontras, Degen, and Goodman (2017) showed that *subjectivity* is a strong predictor of preferred adjective distance from the noun, and subsequent work argued that this regularity follows from restrictive composition and sequential context update: less subjective adjectives contribute more useful information at the inner, more restricted stage of modification (Franke, Scontras, & Simonic, 2019; Scontras, Bar-Sever, Kachakeche, Rosales Jr, & Samonte, 2020; Scontras, Degen, & Goodman, 2019). Separately, Fukumura (2018) demonstrated that speakers in referential tasks tend to place the more *discriminating* adjective first, maximising informativeness in the given context, and Rubio-Fernández, Mollica, and Jara-Ettinger (2021) generalised this into an *incremental efficiency* hypothesis: the utility of a modifier should be evaluated at the point in the utterance where it is encountered, not only after the full description has been heard.

Both lines of work appeal to communicative efficiency, but they locate it at different levels — one in the compositional semantics, the other in the online production process. The central question of this paper is how these two levels interact. Subjectivity provides a stable ordering bias, but the communicative context determines how strongly that bias is expressed in a particular scene. When one property suffices for identification and another does not, the more useful property may exert pressure for earlier mention even when it conflicts with the canonical order. What is missing is a unified account that separates the contribution of stable compositional asymmetries from context-sensitive, online production pressures and tests whether these mechanisms are independently additive or synergistic.

We address three questions. First, can canonical ordering preferences already arise from *context-sensitive restrictive composition* without an incremental production mechanism? Second, does an explicitly *incremental* speaker — one that chooses modifiers word by word — add explanatory value beyond a global, utterance-level evaluation? Third, does *context-updating semantics*, in which the meaning of a gradable adjective is recomputed from the listener's running posterior, provide a further benefit — and is that benefit contingent on incremental production?

To answer these questions, we develop a Rational Speech Act (RSA) model (Degen, 2023; Goodman & Frank, 2016) that crosses two dimensions in a 2×2 comparison: *speaker architecture* (global vs. incremental) and *semantic regime* (context-fixed vs. context-updating). The global speaker evaluates complete utterances; the incremental speaker chooses one modifier at a time. Under context-updating semantics, the interpretation of "big" depends on what the listener already believes after hearing earlier words, so that "big blue" and "blue big" need not be equivalent; under context-fixed semantics, the size threshold is determined once from the scene. All four model variants are fitted to data from two complementary referential experiments in German — a slider-rating study measuring graded ordering preferences and a constrained production study measuring actual utterance choice — in a shared paradigm that varies discriminatory strength across scenes. Model comparison uses Bayesian leave-one-out cross-validation (PSIS-LOO; Vehtari, Gelman, & Gabry, 2017), which quantifies how much predictive accuracy each mechanism adds and detects their interaction. A simulation study complements the empirical work by crossing both speaker architectures with both semantic regimes to identify which ordering effects require context-sensitive composition, which require incremental production, and which emerge only from their interaction.

Both experiments confirm that ordering preferences vary systematically with discriminatory strength: speakers tend to place the more informative adjective earlier while preserving a baseline size-first tendency. The simulation shows that recursive semantics is necessary for the baseline size-first advantage: the global speaker produces the canonical ordering only under context-updating semantics, where threshold adaptation makes the two orderings informationally non-equivalent. In the model comparison, incremental production emerges as the dominant mechanism across both datasets. In the production data, context-updating semantics provides a significant further improvement, but only when paired with an incremental speaker — an interaction indicating that the two mechanisms are synergistic. Figure 1 provides a schematic overview of the modelling framework, experiments, and key findings.

2 Background

Three lines of research bear on the question of how speakers choose and order adjectives, yet to date they have developed largely independently: subjectivity-based accounts trace canonical ordering preferences to compositional interpretation, efficiency-based accounts link adjective choice and order to the discriminatory strength covered by the referential context, and incremental pragmatic models formalize the word-by-word process through which speakers produce utterances. The model developed in this paper integrates all three.

2.1 Subjectivity and Adjective Ordering

A central empirical generalization in the adjective-ordering literature is the relation between an adjective's *subjectivity* and its preferred distance from the noun. Scontras et al.

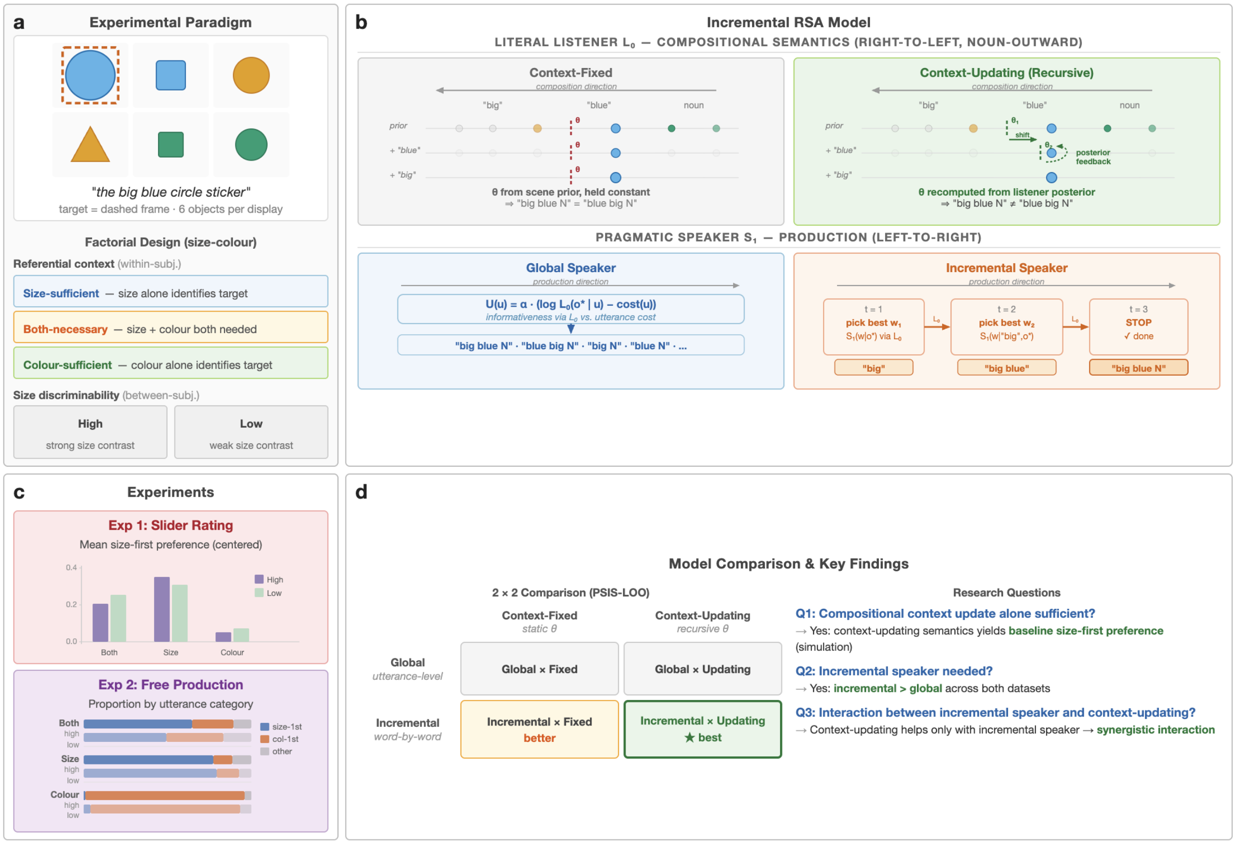

Figure 1

Overview of the present study. (***a***) *Experimental paradigm: six-object displays varying in referential context (within-subj.) and size discriminability (between-subj.).* (***b***) *The RSA model crosses two dimensions: compositional semantics (context-fixed vs. context-updating* θ*, right-to-left) and speaker architecture (global vs. incremental, left-to-right).* (***c***) *Empirical results: slider ratings of ordering preference and production proportions by utterance category.* (***d***) 2×2 *model comparison (PSIS-LOO) and key findings (Q1–Q3).*

(2017) operationalized subjectivity as susceptibility to faultless disagreement and showed that it explains a large share of variance in English ordering preferences, outperforming earlier predictors such as inherentness and intersectivity. Subsequent cross-linguistic work suggests that this relationship is not peculiar to English, but reflects a broader regularity in adjective ordering (Scontras, 2023; Trainin & Shetreet, 2021).

Scontras et al. (2019) argue that the explanatory core of this effect lies in restrictive composition. Adjectives closer to the noun apply over a denotation that has already been narrowed by outer modifiers. In such a restricted comparison class, a less subjective adjective is advantageous because its interpretation is more stable. Franke et al. (2019) make the same intuition explicit in simulation, showing that subjectivity-aligned orders increase average communicative success under sequential composition with noisy meanings. Crucially, this explanation does not require a word-by-word production mechanism; the ordering pressure can emerge from the compositional system itself.

That conclusion is strengthened by Scontras et al. (2020), who argue that what matters is *incremental semantic restriction*. If nominal structure supports sequential narrowing of the comparison class, then subjectivity-based ordering pressures should arise even without an incremental speaker in the usual process-model sense. This identifies a plausible minimal mechanism for canonical ordering: context-sensitive restrictive composition may already suffice to produce a baseline size-first tendency.

2.2 Communicative Efficiency and Discriminatory Strength

The subjectivity account, however, addresses only one source of ordering pressure. A separate line of work has shown that adjective use is sensitive to the demands of the immediate communicative context, including whether a modifier helps the listener identify the referent quickly and efficiently (Degen, Hawkins, Graf, Kreiss, & Goodman, 2020; Rubio-Fernández, 2016; Rubio-Fernandez, 2019). This literature treats adjective choice and adjective order as part of a broader problem of reference generation rather than as the expression of a static lexical hierarchy.

Fukumura (2018) provide a direct precedent for the present experiments. In their production studies, speakers were more likely to place the more discriminating adjective first, consistent with a *discriminatory efficiency hypothesis*: order should reflect which property best distinguishes the target in the current scene. This is a different source of pressure from subjectivity, because it is inherently local. The same adjective may be highly useful in one display and much less useful in another.

Rubio-Fernández and colleagues place this idea in a broader framework of *incremental efficiency* (Rubio-Fernandez & Jara-Ettinger, 2020; Rubio-Fernández et al., 2021). Their central claim is that the efficiency of a referential expression should be evaluated as the utterance unfolds, not only after the full description has been heard. From this perspective, word order matters because it determines how the listener's search space is narrowed over time. Prenominal and postnominal systems therefore create different pragmatic affordances, and even redundant modifiers can be useful if they facilitate early search. What matters is not merely whether a property contributes to unique reference in the end, but whether it has discriminatory value at the relevant stage of processing.

For the present paper, this distinction motivates the central empirical manipulation. Within the modeled size–colour subset, the experiments vary whether size is sufficient, colour is sufficient, or whether both properties are necessary for identification, and they additionally vary how discriminable the size contrast is. These manipulations operationalize the context-sensitive component of the theory: if adjective ordering is responsive to discriminatory strength, then preferences and production should shift with the referential scene rather than reflecting only a global size-first prior.

2.3 Incremental Pragmatic Models

The Rational Speech Act (RSA) framework offers a natural way of formalizing these pressures. In RSA, a pragmatic speaker selects utterances by reasoning about a listener's posterior, and the listener in turn reasons about the speaker's choice (Goodman & Frank, 2016). Standard RSA treats complete utterances as the units of choice, but the same logic can be defined incrementally, with the speaker choosing one word at a time on the basis of how each token updates the listener's beliefs (CohnGordon, Goodman, & Potts, 2019). This extension is relevant to adjective ordering because it makes the utility of a modifier depend on its position in the string.

Existing incremental pragmatic models and related work on redundancy already establish that word-by-word reasoning can capture important asymmetries between adjective positions and between prenominal and postnominal systems (Degen et al., 2020; Rubio-Fernández et al., 2021). What they usually do not assume, however, is that lexical meanings themselves change as the listener posterior changes. In most of these accounts, the posterior updates online but the interpretation of an adjective remains fixed relative to the scene. This amounts to what we will call *context-fixed* semantics: the semantic value of each adjective is determined by the scene-level context and does not depend on which other words have already been processed.

A *context-updating* alternative was introduced by Schlotterbeck and Wang (2023) as part of an incremental RSA model for adjective ordering. Under context-updating semantics, the threshold for gradable adjectives such as size is recomputed from the listener's current posterior at each stage of composition. After hearing "blue," for example, the listener's beliefs concentrate on blue objects, and the comparison class for evaluating "big" narrows accordingly. The result is order-sensitive composition: "blue big" and "big blue" can induce different comparison classes and therefore different communicative outcomes even when they contain the same words. This connects to Teodorescu's (2006) observation that ordering restrictions weaken when alternative orders are not truth-conditionally equivalent — precisely the situation that context-updating semantics creates. By comparing models with context-updating and context-fixed semantics, we can ask not only whether incremental production adds value, but also whether context-updating threshold computation provides a further benefit — and, crucially, whether that benefit is contingent on the speaker architecture.

That question is motivated by a narrower concern than the introduction of incrementality as such. We do not argue that production-level incrementality is required for adjective-ordering preferences in general. The subjectivity literature and, as shown below, our own simulations indicate that context-sensitive restrictive composition already accounts for much of the canonical ordering tendency. The question is instead how much explanatory work remains once that baseline is in place, and whether an incremental production model with context-updating semantics better captures the context-sensitive behavior revealed by the experiments.

The present paper builds directly on Schlotterbeck and Wang (2023), who introduced both the incremental RSA speaker with context-updating semantics and the slider-rating paradigm that forms the basis for one of the experiments reported here. That study demonstrated that the model qualitatively captures a subjectivity-driven baseline ordering as well as a discriminatory-strength modulation, but it did not isolate the respective contributions of incremental production and context-updating semantics, nor did it assess their quantitative fit. We extend that work in four directions: we embed the model in a systematic 2×2 comparison of speaker architecture and semantic regime, including a context-fixed ablation; we add a constrained production experiment that provides a richer response space; we introduce a simulation study that identifies which ordering effects follow from context-updating composition alone; and we use Bayesian model comparison to quantify each mechanism's contribution and to reveal an interaction between incremental production and context-updating semantics that was not visible in the earlier qualitative analysis.

3 Computational Model

This section develops the RSA model used in the present paper. The model builds on the standard RSA architecture (Goodman & Frank, 2016): a literal listener interprets utterances using a compositional semantics, and a pragmatic speaker chooses utterances by reasoning about the listener's posterior belief. Following Schlotterbeck and Wang (2023), the model incorporates two extensions of the standard architecture: an incremental speaker that chooses one word at a time, and a context-updating compositional semantics in which the meaning of gradable adjectives depends on the listener's running posterior. The experiments were broader than the fitted model comparison: the full paradigm included size–form and colour–form trials, but the analyses reported here focus on size–colour scenes, where the theoretical contrast between a gradable adjective and a determinate feature adjective is clearest. Form therefore remains part of the general representational vocabulary and of the production response space, but it is not the primary mechanistic contrast in the current fit.

3.1 Representational Primitives

An object is a triple $o = (s, c, f)$, where $s \in [1, 10]$ denotes size, $c \in \{0, 1\}$ indicates colour match, and $f \in \{0, 1\}$ indicates form match. A context $\mathcal{C}$ is a set of n_{obj} objects. In the experiments, the intended referent is always the largest object with $c = f = 1$.

Utterances are built from three adjective types, S, C, and F. An utterance $u = (a_1, \ldots, a_T)$ is an ordered sequence of adjectives $a_t \in \{\mathrm{S}, \mathrm{C}, \mathrm{F}\}$ with $T \in \{1, 2, 3\}$. Combining all lengths and orderings yields 15 surface forms: {S, SC, SCF, SF, SFC, C, CS, CSF, CF, CFS, F, FS, FSC, FC, FCS}. This full inventory reflects the broader experimental vocabulary. For the reported model fits, however, the scene subset is size–colour: C denotes the determinate feature that competes with size for identification, while F remains available chiefly as an additional, potentially redundant modifier in production.

3.2 Meaning Functions

Context-dependent size semantics. We model size adjectives with a context-sensitive threshold semantics following Schmidt, Goodman, Barner, and Tenenbaum (2009), who showed that the interpretation of gradable adjectives depends on the statistics of the comparison class. Franke et al. (2019) adopted the same approach in their simulation of adjective-ordering preferences. A threshold is computed from the current comparison class $\mathcal{C}$:

$$\theta_k(\mathcal{C}) = x_{\max}(\mathcal{C}) - k\left(x_{\max}(\mathcal{C}) - x_{\min}(\mathcal{C})\right), \tag{1}$$

where $x_{\min}(\mathcal{C})$ and $x_{\max}(\mathcal{C})$ are the lower and upper bounds of the mid-range of the contextual size distribution and $k \in [0, 1]$ controls how permissively "big" is applied. The resulting semantic value $[\![\mathrm{S}]\!](o, \mathcal{C}) \in [0, 1]$ is graded rather than binary: an object's size is compared to the threshold via a normal slack function scaled by a perceptual-blur parameter w_f, consistent with scalar variability in the Approximate Number System (Feigenson, Dehaene, & Spelke, 2004). This context dependence is what makes the context-fixed vs. context-updating distinction possible: because θ_k depends on $\mathcal{C}$, changing which objects are in the comparison class changes what counts as big.

Colour and form semantics. Colour and form are in principle determinate: an object either matches the feature or it does not. Following the relaxed-semantics approach of Degen et al. (2020) and Waldon and Degen (2021), however, we soften these denotations by parameterizing them with $\nu \in (0, 1)$, which can be interpreted as the probability that a feature-matching object is correctly described by the adjective. This relaxation allows even determinate properties to carry graded information, which is important for modeling redundant modification. In the reported model fits, colour is the primary determinate comparison case because inference is carried out on size–colour scenes, whereas form is retained as a general determinate feature term and as a possible redundant modifier in production. In the slider models, colour and form share a single ν parameter. In the production models, they receive separate fixed values, ν_{colour} and ν_{form}, obtained from a preliminary inference run and held constant across all model variants.

3.3 Incremental Listener

The literal listener constructs utterance meanings incrementally by combining token-level semantic contributions while updating the effective comparison class. Let $u = (w_1, \ldots, w_T)$ be an utterance. The listener maintains a sequence of belief states $\{P_t(o)\}_{t=0}^{T}$, where $P_t(o)$ denotes the posterior belief after incorporating the suffix $(w_{t+1}, \ldots, w_T)$:

$$P_t(o) = P\Big(o \mid w_{t+1}, \ldots, w_T\Big), \qquad t = 0, \ldots, T, \tag{2}$$

with base case

$$P_T(o) = P_{\text{prior}}(o). \tag{3}$$

For $t = T, T-1, \ldots, 1$, beliefs are updated according to

$$P_{t-1}(o) \propto P_t(o) \, [\![w_t]\!](o), \tag{4}$$

which normalizes to

$$P_{t-1}(o) = \frac{P_t(o) \, [\![w_t]\!](o)}{\sum_{o'} P_t(o') \, [\![w_t]\!](o')}. \tag{5}$$

After all tokens have been processed, the listener arrives at

$$P_0(o) = P(o \mid u). \tag{6}$$

The composition rule in Equation (4) is shared by both semantic regimes of the model. What differs is how the meaning function $[\![w_t]\!]$ is computed for gradable adjectives such as size.

Context-updating semantics. Under the context-updating regime, the size threshold θ_k is recomputed from the listener's current belief state at each composition step:

$$[\![w_t]\!]_{\text{UPD}}(o) \;=\; [\![w_t]\!]_{P_t}(o), \tag{7}$$

where P_t is the posterior after processing all words to the right of position t. The current belief state therefore plays a dual role: it serves as the prior for the next update step, and it determines the semantics of the next token. Because P_t itself depends on the semantics of later words, this creates a feedback loop:

$$[\![w_t]\!]_{P_t} \quad \text{depends on} \quad P_t(o) \propto P_{t+1}(o) \, [\![w_{t+1}]\!]_{P_{t+1}}(o). \tag{8}$$

Word order changes the sequence of comparison classes and thereby the interpretation of the utterance, yielding non-commutativity:

$$P(o \mid w_1, w_2) \neq P(o \mid w_2, w_1) \quad \text{in general.} \tag{9}$$

Context-fixed semantics. Under the context-fixed regime, the size threshold is computed once from the scene prior and held constant throughout composition:

$$[\![w_t]\!]_{\text{FIX}}(o) \;=\; [\![w_t]\!]_{P_{\text{prior}}}(o) \quad \text{for all } t. \tag{10}$$

The listener's beliefs still update incrementally via Equation (4), but this update no longer feeds back into the meaning function. For deterministic adjectives such as colour and form, both regimes are identical because their semantics does not depend on the comparison class. The context-fixed regime therefore isolates the contribution of incremental belief updating from the additional effect of posterior-dependent meaning.

3.4 Utterance Prior and Speaker Models

The model incorporates a prior over utterance forms that is separate from the pragmatic component. This prior captures stable ordering tendencies — such as the well-documented subjectivity-driven size-first preference — that the speaker brings to the task before any scene-specific pragmatic reasoning.

In the slider-rating models, the prior is minimal: a single inferred bias parameter $b \geq 0$ that penalizes the reverse (colour-first) order relative to the canonical (size-first) order. Because b is estimated from data rather than fixed, the posterior indicates how much residual ordering preference remains after the RSA mechanism has been applied; if the pragmatic model were sufficient on its own, the posterior for b would concentrate near zero.

In the production models, the prior is richer: a language-model distribution $P_{\text{LM}}(u)$ over the 15 utterance forms, derived from German GPT-2 surprisal estimates averaged across the experimental adjective–noun combinations. The contribution of this prior is scaled by an inferred parameter β, yielding utterance cost $\text{cost}_{\text{LM}}(u) = -\beta \log P_{\text{LM}}(u)$. Like b in the slider model, the LM prior encodes stable word-order tendencies from language experience, which the pragmatic speaker then modulates in light of the current scene.

Global speaker. The global speaker computes a distribution over complete utterances by applying a softmax to utterance-level utility:

$$S_{\text{gb}}(u \mid o) = \frac{\exp\Big(U(o,u)\Big)}{\sum_{u'} \exp\Big(U(o,u')\Big)}. \tag{11}$$

For the production model, utility is defined as

$$U(o^*, u) = \alpha \, \log L(o^* \mid u) + \beta \, \log P_{\text{LM}}(u), \tag{12}$$

where α is the RSA rationality parameter. In the slider-rating models, the utility simplifies to

$$U(o,u) = \alpha\Big(\log L(o \mid u) - b_u\Big), \tag{13}$$

where $b_u = 0$ for the size-first order and $b_u = b$ for the reverse order.

Incremental speaker. The incremental speaker chooses one token at a time. For an intended referent o^*,

$$S_{\text{inc}}(u \mid o^*) = \prod_{t=1}^{T} S_t(w_t \mid u_{<t}, o^*)\,, \qquad u_{<t} = (w_1, \ldots, w_{t-1}), \tag{14}$$

with local decision rule

$$S_t(a \mid u_{<t}, o^*) \propto \Big[L(o^* \mid u_{<t} + a)\Big]^{\alpha}. \tag{15}$$

Combining local pragmatic choice with utterance-level cost yields, for the production model,

$$S_{\text{inc}}(u \mid o^*) \propto P_{\text{LM}}(u)^{\beta} \cdot \prod_{t=1}^{T} \frac{\Big[L(o^* \mid u_{<t} + w_t)\Big]^{\alpha}}{Z_t(u_{<t}, o^*)}, \tag{16}$$

where

$$Z_t(u_{<t}, o^*) = \sum_a \Big[L(o^* \mid u_{<t} + a)\Big]^{\alpha} \tag{17}$$

ensures local normalization. In the slider-rating models, the LM prior is replaced by the bias term: $S_{\text{inc}}(u \mid o^*) \propto \exp(-b_u) \cdot \prod_t S_t(w_t \mid u_{<t}, o^*)$. The theoretical contrast is that the global speaker applies pragmatic reasoning once over complete utterances, whereas the incremental speaker applies it at each token position. When semantics is context-dependent, these two model variants are not equivalent.

3.5 Model Variants and Inference

We report a 2×2 model comparison that crosses two theoretically motivated dimensions: *speaker architecture* (global vs. incremental) and *semantic regime* (context-fixed vs. context-updating). All four variants use hierarchical participant pooling, which consistently outperforms pooled alternatives (see Appendix). The speaker-architecture contrast asks whether word-by-word production planning adds explanatory value beyond utterance-level evaluation. The semantic-regime contrast asks whether recomputing the size threshold from the listener's evolving belief state adds value beyond a fixed comparison class. Their interaction tests whether these two mechanisms are synergistic.

The two semantic regimes — context-updating (Equation (7)) and context-fixed (Equation (10)) — are defined formally above. All other aspects of the model, including the belief-update structure and the speaker's decision rule, remain identical across regimes. The context-fixed variant therefore serves as a targeted ablation of the context-updating mechanism.

The semantic parameters $k = 0.5$, $w_f = 1.0$, $\nu_{\text{colour}} = 0.85$, and $\nu_{\text{form}} = 0.70$ (production only) are fixed across all variants. Their influence on model predictions is explored in the simulation study below.

All models were implemented in NumPyro (Phan, Pradhan, & Jankowiak, 2019) and estimated with the NUTS sampler (Hoffman & Gelman, 2014). Convergence was assessed via split-$\hat{R}$, effective sample size, divergence counts, and posterior predictive checks.

The two datasets required different likelihood specifications. Slider responses showed substantial mass at the boundaries (0 and 1), which motivated a Zero-One-Inflated Beta (ZOIB) likelihood rather than a Gaussian approximation. The free-production data were categorical and highly skewed, with many utterance types near zero frequency in some conditions. To keep inference stable and the comparison interpretable, the semantic parameters were fixed across all variants and only the pragmatic parameters were estimated from data. This ensures that any difference in fit between variants reflects the speaker

Table 1

Slider-rating model variants (2 × 2: speaker × semantics, all hierarchical)

Model	Speaker	Size semantics	Inferred parameters
Global, context-fixed	S_{G}	Fixed threshold from scene prior	α, b, σ, τ, δ_i
Global, context-updating	S_{G}	Threshold updated from listener posterior	α, b, σ, τ, δ_i
Incremental, context-fixed	S_{I}	Fixed threshold from scene prior	α, b, σ, τ, δ_i
Incremental, context-updating	S_{I}	Threshold updated from listener posterior	α, b, σ, τ, δ_i

architecture and semantic regime, not incidental variation in lexical semantics.

4 Experiments

We tested the model predictions with two complementary online referential tasks in German. Experiment 1 used graded slider judgments to measure how acceptable competing adjective orders are in a controlled scene. Experiment 2 used constrained production to measure which adjective sequences speakers actually choose in the same referential displays. Combining the two tasks allows us to separate stable ordering preferences from their online realization in production.

4.1 Shared Method

4.1.1 Design

Both experiments used the same referential design. On every critical trial, participants saw six objects and had to refer to the framed target by its perceptual

Table 2

Free-production model variants (2 × 2: speaker × semantics, all hierarchical)

Model	**Speaker**	**Size semantics**	**Inferred parameters**
Global, context-fixed	S_{G}	Fixed threshold from scene prior	α, β, τ, δ_i
Global, context-updating	S_{G}	Threshold updated from listener posterior	α, β, τ, δ_i
Incremental, context-fixed	S_{I}	Fixed threshold from scene prior	α, β, τ, δ_i
Incremental, context-updating	S_{I}	Threshold updated from listener posterior	α, β, τ, δ_i

properties rather than by screen position. The target was always the largest object that matched the intended colour and form. The analyses reported here focus on the theoretically central size–colour subset. Within that subset, referential context had three levels: size-sufficient (size alone identifies the target), colour-sufficient (colour alone identifies the target), and both-necessary (neither property alone suffices). A second factor, size discriminability, varied whether the size contrast between the target and its competitors was high or low. Referential context varied within participants; size discriminability was manipulated between participants.

The broader paradigm also included size–form and colour–form trials to widen the material base and reduce dependence on a single adjective pairing. These conditions remain part of the experimental design, but the present analysis concentrates on the size–colour subset because it provides the clearest contrast between a gradable adjective and a determinate feature adjective.

Table 3

Shared design of the reported size–colour analyses

Factor	Levels
Referential context	Size-sufficient, colour-sufficient, both-necessary
Size discriminability	High (strong size contrast), low (weak size contrast)

Fixed display property. The referent is the largest, colour-matching, form-matching object in a six-object context.

4.1.2 Materials

The experiments were built from 27 underlying scene types. Each display contained six objects varying in size and in discrete colour and form values. The target was marked with a coloured frame. Critical vocabularies were fixed across the study, with six colour terms and six form terms reused throughout the scene set, and the modified noun was always *Aufkleber* ('sticker'). Items were distributed across participants in a Latin-square design so that each participant saw every scene type exactly once in each condition, and condition–item pairings were counterbalanced across participants. In the computational model, each display can be represented as a 6×3 matrix (s, c, f) with the referent at index 0, but the experimental task presented these scenes as ordinary visual arrays rather than as explicit feature lists.

4.1.3 Procedure

Participants completed the experiments online via Prolific and were instructed to identify the framed target using its perceptual properties, not spatial descriptions such as *left* or *right*. The cover story framed each trial as a telephone conversation about collectible stickers: participants saw a set of six stickers on screen, one highlighted by a coloured frame, and were asked to refer to the target for a partner who could see the same stickers but neither the frame nor the spatial layout. This made spatial terms uninformative and forced reliance

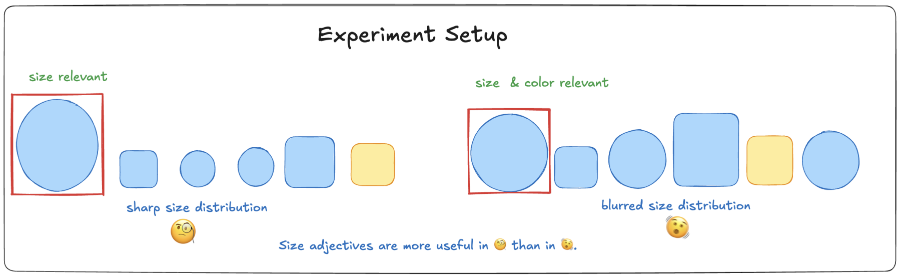

Figure 2

Illustration of an experimental item. The referent (circled) is the largest, colour-matching, form-matching object in a six-object display. The two panels illustrate high- and low-discriminability size conditions.

on adjective descriptions. Both experiments began with a consent form and a brief practice block (three training trials with feedback) that familiarised participants with the interface and the referential task. The main trials were divided into four blocks of approximately equal length, with self-paced breaks between blocks. Trial order within each block was randomised independently for each participant. The two experiments differed in response format: Experiment 1 presented two competing adjective orders on a continuous slider (180 trials total, drawn from one of six counterbalanced lists), whereas Experiment 2 asked participants to type the adjective material they would place before the noun (135 trials, again from one of six lists, plus two additional dictionary-style practice items). Despite this difference, both tasks probed the same question: whether adjective ordering tracks baseline size-first preferences, referential context, and the discriminatory value of size in the current display.

4.2 Experiment 1: Slider-Rating Study

4.2.1 Participants

Participants ($N = 134$ after exclusions; 70 low discriminability, 64 high discriminability) were recruited online via Prolific and reported German as their native or fluent language. Exclusion criteria were: more than three missing trial-list entries, completion times more than two standard deviations from the sample mean, more than three failures on control items, or more than three “none fits” responses on experimental items.

The analysed dataset contained 10,606 critical observations. Mean completion time was 27.6 minutes. Participants were compensated via Prolific at a rate above the platform-recommended minimum.

4.2.2 Task and Scoring

Experiment 1 implemented the shared referential design as a comparative judgment task. Participants were asked to imagine a referential exchange about sticker collections and to evaluate which description would be better for identifying the framed target. On each critical trial, two competing adjective orders appeared on opposite sides of a continuous slider, together with a “none of the descriptions fits” option. Left–right presentation of the two orders was randomised across trials. Each participant saw one of six lists built from the 27 underlying scenes, yielding 81 critical trials and 99 filler and control trials (180 total). The filler and control items discouraged fixed response strategies and ensured attention to the referential task. Raw slider values were recorded on a 0–100 scale and rescaled to $[-0.5, 0.5]$, with positive values indicating a size-first preference. The descriptive analysis used a linear mixed-effects model with centred slider rating as the dependent variable and referential context, size discriminability, and their interaction as fixed effects, with random intercepts for participant and item. For the Bayesian model comparison, these ratings were modelled with a Zero-One-Inflated Beta (ZOIB) likelihood to accommodate boundary mass at 0 and 1.

4.2.3 Results

Figure 3 shows that size-first orderings were preferred across all three referential contexts, but the strength of the preference depended on which property was informative. Referential context produced a large main effect, $F(2, 3349) = 213.7$, $p < .001$; $\chi^2(2) = 402.6$, $p < .001$. The size-first preference was strongest in size-sufficient displays, intermediate in both-necessary displays, and weakest in colour-sufficient displays. Notably, even colour-sufficient contexts did not produce a qualitative reversal: mean centred ratings remained above zero.

Size discriminability did not produce a reliable main effect, $F(1, 132) = 0.03$, $p = .86$, but it interacted with referential context, $F(2, 3349) = 5.84$, $p = .003$; $\chi^2(2) = 11.7$, $p = .003$. The interaction was asymmetric: higher discriminability slightly increased the size-first preference in size-sufficient contexts (low: 0.30, high: 0.34) but slightly reduced it in both-necessary (0.25 vs. 0.20) and colour-sufficient contexts (0.06 vs. 0.04). Post-hoc pairwise comparisons (Bonferroni-corrected) confirmed that no individual simple effect of discriminability within any referential context reached significance (all $p > .12$), indicating that the interaction is driven by the directional pattern across contexts rather than by any single cell contrast (see Appendix for full pairwise results). This pattern is consistent with a discriminatory-strength interpretation: ordering preferences track the usefulness of size for identifying the target, but a baseline size-first tendency persists even when colour is locally more discriminating.

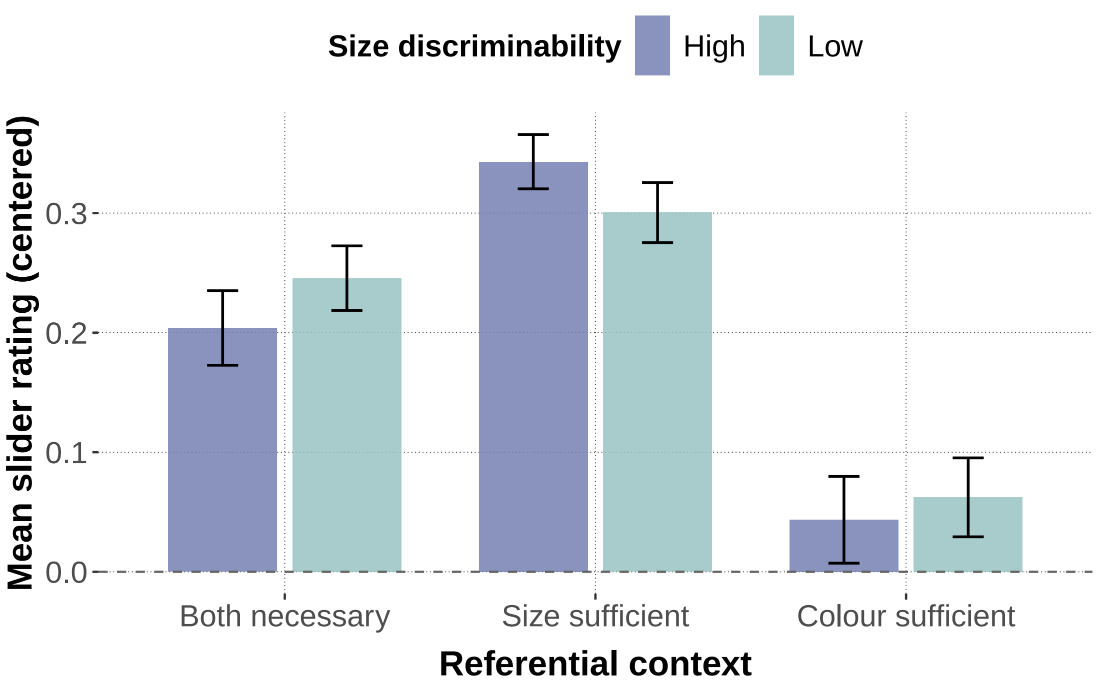

Figure 3

Empirical results from the slider-rating study. Mean centered slider ratings are shown by referential context and size discriminability, with 95% confidence intervals. Positive values indicate a preference for size-first orderings.

4.3 Experiment 2: Free-Production Study

4.3.1 Participants

Participants ($N = 113$ after exclusions; 60 low discriminability, 53 high discriminability) were recruited online via Prolific; anyone who had completed the slider study was excluded. Further exclusion criteria were: completion times more than two standard deviations from the mean, more than 80% identical responses, or more than 10% missing or unannotatable responses. The analysed dataset contained 9,639 critical observations. Mean completion time was 32.4 minutes. Compensation followed the same Prolific rate as Experiment 1.

4.3.2 Task and Coding

Experiment 2 used the shared referential design in a constrained production format. Participants saw the framed target together with a sentence prompt and typed the adjective material needed before the noun *Aufkleber*. Restricting the response to the adjective slot rather than eliciting a full sentence reduced off-task variability and provided a controlled basis for coding. Instructions again prohibited spatial descriptions, and a practice block familiarised participants with the relevant colour and form vocabulary before the critical trials began. Each list contained 81 critical trials; the filler load was reduced relative to Experiment 1 (135 total trials vs. 180) to accommodate the greater time burden of typed production.

Responses were coded into 15 utterance categories corresponding to all one-, two-, and three-adjective strings built from size, colour, and form terms (3 one-adjective, 6 two-adjective, 6 three-adjective). Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous; spatial descriptions, comparative or ordinal size expressions, and other unclassifiable responses were excluded. The dependent variable for the Bayesian model comparison is the categorical utterance label $y_i \in \{1, \ldots, 15\}$ for each trial i. For descriptive statistics and targeted mixed-effects analyses, we also tracked two derived binary outcomes: whether an utterance was size-initial and whether it was over-informative (i.e., a three-adjective response in both-necessary contexts, or a multi-adjective response in size-sufficient or colour-sufficient contexts).

4.3.3 Results

The production data show stronger context sensitivity than the slider ratings. As Figure 4 illustrates, size-first utterances dominated in size-sufficient contexts, colour-first utterances dominated in colour-sufficient contexts, and both-necessary contexts fell between the two extremes. A mixed-effects logistic regression on size-initial responses yielded a strong main effect of referential context, $\chi^2(2) = 1859.7$, $p < .001$, no reliable main effect of size discriminability, $\chi^2(1) = 0.79$, $p = .37$, and a reliable interaction, $\chi^2(2) = 31.4$, $p < .001$. As in the slider data, the interaction was asymmetric: higher discriminability reduced the size-first advantage in size-sufficient contexts but increased it in both-necessary contexts.

A complementary analysis of over-informativity reveals a related efficiency pressure. Over-informative utterances were most frequent in size-sufficient contexts and least frequent in colour-sufficient contexts, producing a strong main effect of referential context, $\chi^2(2) = 725.7$, $p < .001$. Unlike size-initiality, over-informativity also showed a main effect of size discriminability, $\chi^2(1) = 10.3$, $p = .001$, with more multi-adjective responses in low- than in high-discriminability displays. This effect interacted with referential context, $\chi^2(2) = 63.8$, $p < .001$, driven primarily by size-sufficient contexts. The pattern aligns with the broader communicative-efficiency literature on redundancy: speakers produce additional material when it helps resolve reference, even if it is not strictly necessary for uniqueness.

4.4 Interim Summary

Across both experiments, three empirical regularities emerge. First, there is a robust baseline preference for size-first orderings that persists even when colour is the only discriminating property. Second, this preference is modulated by referential context: size-first choices increase when size is informative and decrease when colour is more useful, consistent with a discriminatory-efficiency account. Third, size discriminability interacts with referential context but does not produce reliable main effects, suggesting that the

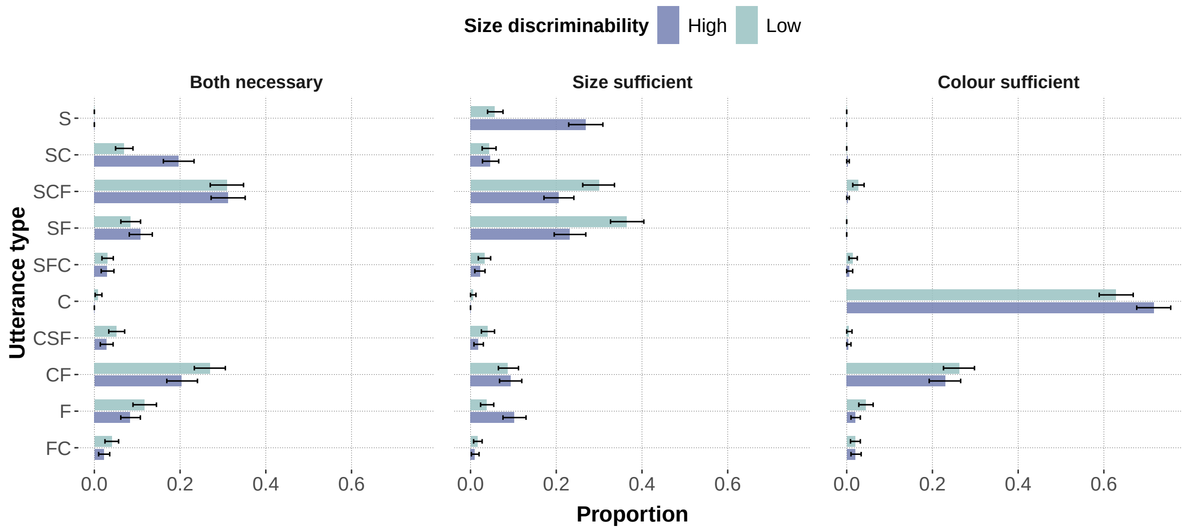

Figure 4

Empirical results from the free-production study. Proportions of the major utterance types are shown by referential context and size discriminability, with bootstrapped 95% confidence intervals. Only utterance types reaching at least 2% in one condition are displayed.

sharpness of the size contrast acts as a secondary modulator rather than a primary driver. The production data amplify these patterns relative to the slider data: free production elicits stronger context sensitivity and reveals a complementary over-informativity gradient driven by the same referential pressures. Together, these results motivate a formal model that can capture both the stable ordering bias and its context-dependent modulation, while accommodating the richer response space of unconstrained production.

5 Bayesian Model Comparison

We evaluated the four model variants (Tables 1 and 2) against both datasets using Pareto-smoothed importance-sampling leave-one-out cross-validation (PSIS-LOO; Vehtari et al., 2017). All models were estimated with four MCMC chains using the No-U-Turn Sampler (NUTS; Hoffman & Gelman, 2014) as implemented in NumPyro (Phan et al., 2019). For the slider models, each chain drew 500 warmup and 500 post-warmup samples (target acceptance probability = 0.90, maximum tree depth = 8). For the production models, each chain drew

5,000 warmup and 2,000 post-warmup samples (target acceptance probability = 0.85, maximum tree depth = 7). Convergence was confirmed via split-$\hat{R} < 1.01$ and effective sample sizes exceeding 400 for all parameters across all models. PSIS-LOO comparisons are reported as the difference in expected log pointwise predictive density (ΔELPD) relative to the best-fitting model, together with the difference standard error (dse). A ratio $|\Delta\text{ELPD}|/\text{dse}$ substantially greater than 1 indicates a significant difference in out-of-sample predictive accuracy. Table 4 provides the full PSIS-LOO comparison for all eight models.

Table 4

PSIS-LOO comparison for all models. $ELPD_{LOO}$ is the absolute expected log pointwise predictive density; ΔELPD and dse are the difference and its standard error relative to the best-fitting model in each dataset; p_{LOO} is the effective number of parameters.

Dataset	Model	ELPD_{LOO}	ΔELPD	dse	p_{LOO}
Slider-Rating Study					
	Incremental, context-updating	−2,774.1	0.0	—	64.8
	Incremental, context-fixed	−2,774.1	−0.1	0.2	65.1
	Global, context-updating	−4,249.2	−1,475.1	65.9	1,425.0
	Global, context-fixed	−7,562.2	−4,788.1	127.2	4,372.0
Free-Production Study					
	Incremental, context-updating	−6,985.3	0.0	—	69.3
	Incremental, context-fixed	−7,000.9	−15.6	1.5	70.7
	Global, context-updating	−7,728.1	−742.8	45.0	22.9
	Global, context-fixed	−7,738.5	−753.2	44.8	21.1

5.1 Slider-Rating Study

Figure 5 shows the 2×2 comparison for the slider data. The dominant contrast is speaker architecture: both incremental models fit the data well and are essentially indistinguishable ($|\Delta\text{ELPD}|/\text{dse} = 0.32$), whereas both global models perform far worse

(ΔELPD $> 1{,}400$). Within the incremental pair, context-updating semantics confers no detectable advantage over context-fixed semantics. For graded preference judgments between two competing orders, word-by-word pragmatic reasoning is therefore the dominant mechanism: it provides strong order sensitivity regardless of whether the size threshold is recomputed from the running posterior. Posterior-predictive correlations confirm that both incremental models capture the condition-level pattern equally well (see Figure C1 in the Appendix). For the best-fitting incremental context-updating model, the posterior rationality parameter was $\alpha = 0.69$ ($SD = 0.06$, 94% HDI $[0.57, 0.79]$), the ZOIB dispersion was $\sigma = 0.72$ ($SD = 0.01$, 94% HDI $[0.70, 0.74]$), and the hierarchical spread of participant-level rationality offsets was $\tau = 0.12$ ($SD = 0.01$, 94% HDI $[0.10, 0.14]$). Crucially, the bias intercept was $b = 0.58$ ($SD = 0.06$, 94% HDI $[0.47, 0.69]$), reliably above 0.50. This positive bias means that even after the RSA mechanism accounts for context-dependent informativeness, the model infers a residual size-first preference — consistent with the view that a stable ordering default persists beyond what discriminatory utility alone explains. Within the global pair, context-updating semantics substantially outperforms context-fixed semantics (ΔELPD $\approx 3{,}313$, $|\Delta\text{ELPD}|/\text{dse} > 24$). Context-updating composition thus provides a minimal source of order sensitivity that partially compensates for the lack of incrementality, but both global models remain far worse than either incremental variant.

5.2 Free-Production Study

The production comparison (Figure 6) reveals a richer pattern. As in the slider study, both incremental models outperform both global models by a wide margin (ΔELPD > 740, $|\Delta\text{ELPD}|/\text{dse} > 16$). Unlike the slider study, however, the semantic-regime contrast now matters within the incremental pair: the incremental context-updating model is clearly the best-fitting variant ($\Delta\text{ELPD} = -15.6$, $|\Delta\text{ELPD}|/\text{dse} = 10.1$, $p < .001$, relative to incremental context-fixed). Within the global pair, the two semantic regimes are effectively tied (ΔELPD ≈ 10).

This interaction between speaker architecture and semantic regime is the most

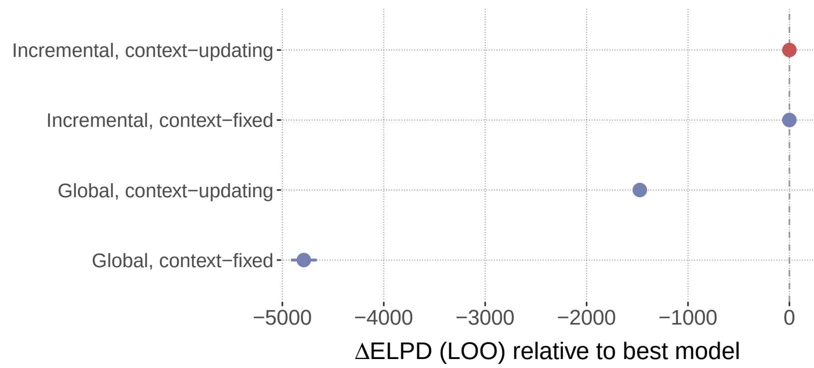

Figure 5

Slider-rating model comparison (2 × 2: speaker × semantics). ΔELPD relative to the best model, with ±1 difference standard error. Models whose intervals exclude zero show significantly worse predictive performance.

theoretically informative result of the model comparison. Context-updating threshold computation adds significant predictive value for production, but only when paired with an incremental decision architecture that can exploit non-commutative composition at each step. The production task is thus precisely the setting where word-by-word planning and context-updating semantics jointly matter, consistent with the view that adjective ordering in production reflects online discriminatory utility rather than utterance-level well-formedness alone. Posterior-predictive correlations for the best-fitting model are shown in the Appendix. For the best-fitting incremental context-updating model, the posterior rationality parameter was $\alpha = 2.09$ ($SD = 0.09$, 94% HDI $[1.92, 2.25]$), the log utterance-prior weight was $\log \beta = -0.24$ ($SD = 0.03$, 94% HDI $[-0.31, -0.18]$), and the hierarchical spread was $\tau = 0.72$ ($SD = 0.07$, 94% HDI $[0.59, 0.84]$). The negative $\log \beta$ (i.e., $\beta \approx 0.79$) indicates a moderate down-weighting of the corpus-derived prior relative to the RSA component. Comparing across model variants, the global models yielded lower α posteriors ($\alpha \approx 1.27$) and less negative $\log \beta$ (≈ -0.17, i.e., $\beta \approx 0.85$), suggesting that the global architecture relies more on the LM prior and less on pragmatic reasoning to fit the data. The substantially larger τ in the incremental models (0.71–0.72 vs. 0.32–0.33) further indicates greater participant-level variability in pragmatic engagement under the incremental architecture.

Three additional production ablation models are reported in full in the Appendix. A lookahead variant, which evaluates both remaining adjective orders before committing to a word, performed worse than the standard greedy incremental model, indicating that local step-by-step planning suffices for this task. An LM-only ablation, which removes the RSA component and relies solely on the corpus-derived prior, likewise fell short, confirming that the utterance prior alone cannot account for context-sensitive ordering. An RSA-only ablation, which removes the LM prior, failed the PSIS-LOO diagnostic entirely (high Pareto-k values), suggesting that purely pragmatic choice without a lexical prior produces degenerate predictive distributions for this response space.

5.3 Cross-Dataset Summary

Comparing the two datasets reveals a diagnostic interaction between speaker architecture and semantic regime. Within the incremental models, context-updating semantics significantly improves fit for production ($|\Delta\text{ELPD}|/\text{dse} = 10.1$) but not for slider ratings ($|\Delta\text{ELPD}|/\text{dse} = 0.32$). Within the global models, the complementary pattern holds: context-updating semantics provides a massive advantage in the slider data ($\Delta\text{ELPD} \approx 3{,}313$) but confers no detectable benefit in production ($\Delta\text{ELPD} \approx 10$). This cross-pattern suggests that context-updating composition is most useful when the model otherwise lacks a source of order-relevant information: in the global slider case, recursive threshold updating is the only mechanism that can break the symmetry between orders; in production, the LM prior already provides ordering information, making the global recursive advantage redundant. Conversely, when the speaker architecture itself introduces order sensitivity through incremental decision-making, context-updating semantics adds value only when the response space is rich enough (15-way production) to reward finer-grained discriminatory computation.

All models reported in the main text use hierarchical participant-level pooling (random intercepts on the rationality parameter). Pooled variants, which estimate a single set of parameters across all participants, were consistently worse and are reported in the Appendix.

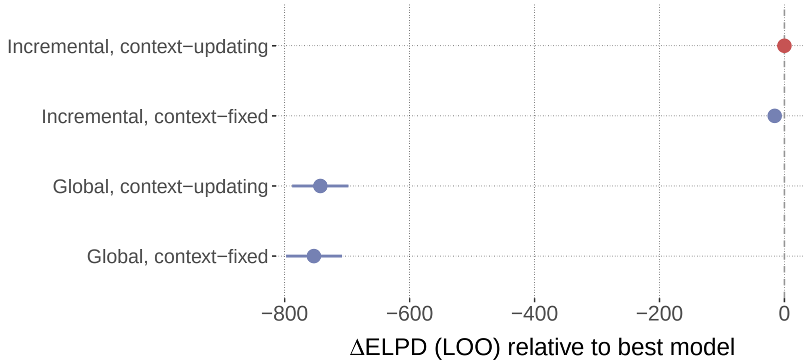

Figure 6

Production model comparison (2×2*: speaker* $\times$ *semantics).* Δ*ELPD relative to the best model, with* ± 1 *difference standard error. Models whose intervals exclude zero show significantly worse predictive performance.*

6 Simulation Study

6.1 Goals and Design

The simulation addresses three questions that the model comparison alone cannot answer. First, does the model *generate* a size-first ordering preference across a broad range of contexts, or does this pattern emerge only after fitting to data? Second, which parameter regions produce a size-first advantage and which, if any, produce a colour-first reversal? Third, how do speaker architecture and semantic regime jointly determine the ordering preference — in particular, is context-updating semantics necessary for the baseline effect, or does it emerge from incremental production alone?

To answer these questions, we ran a full factorial sweep over the parameters listed in Table 5, crossing both speaker models (incremental, global) with both semantic regimes (recursive, static). Under recursive semantics, the size threshold adapts to the literal listener's running posterior at each composition step; under static semantics, the threshold is computed once from the uniform prior and held fixed. For each parameter combination, 1,000 random six-object contexts were sampled, yielding 12,000,000 simulated trials in total. On each trial, the simulation computed the speaker's probability of producing the size-first vs. colour-first ordering for the target referent and recorded the size-first advantage (the difference in communicative success between the two orders).

Table 5

Simulation parameter grid

Parameter	**Values swept**	**# levels**
Speaker model	Incremental, global	2
Semantics	Recursive, static	2
n_{obj}	2, 6, 10, 14, 18, 22, 26, 30	8
σ_{spread}	2.0, 7.75, 15.0	3
ν	0.90, 0.92, 0.94, 0.96, 0.98	5
k	0.10, 0.30, 0.50, 0.70, 0.90	5
w_f	0.1, 0.2, 0.3, 0.5, 0.8	5
Random context samples per combination	1,000	
Total trials	**12,000,000**	

The parameter σ_{spread} controls the spread of the size distribution in the sampled context: low values produce low-discriminability contexts in which size is weakly informative, whereas high values produce high-discriminability contexts in which size strongly separates the target from its competitors. The remaining parameters (k, w_f, ν) are the fixed semantic parameters from the fitted models; their individual effects are reported in the Appendix.

6.2 Results

Figure 7 shows the size-first advantage as a function of context size and contextual spread, faceted by speaker model (columns) and semantic regime (rows).

Three findings stand out. First, under recursive semantics, both speaker models

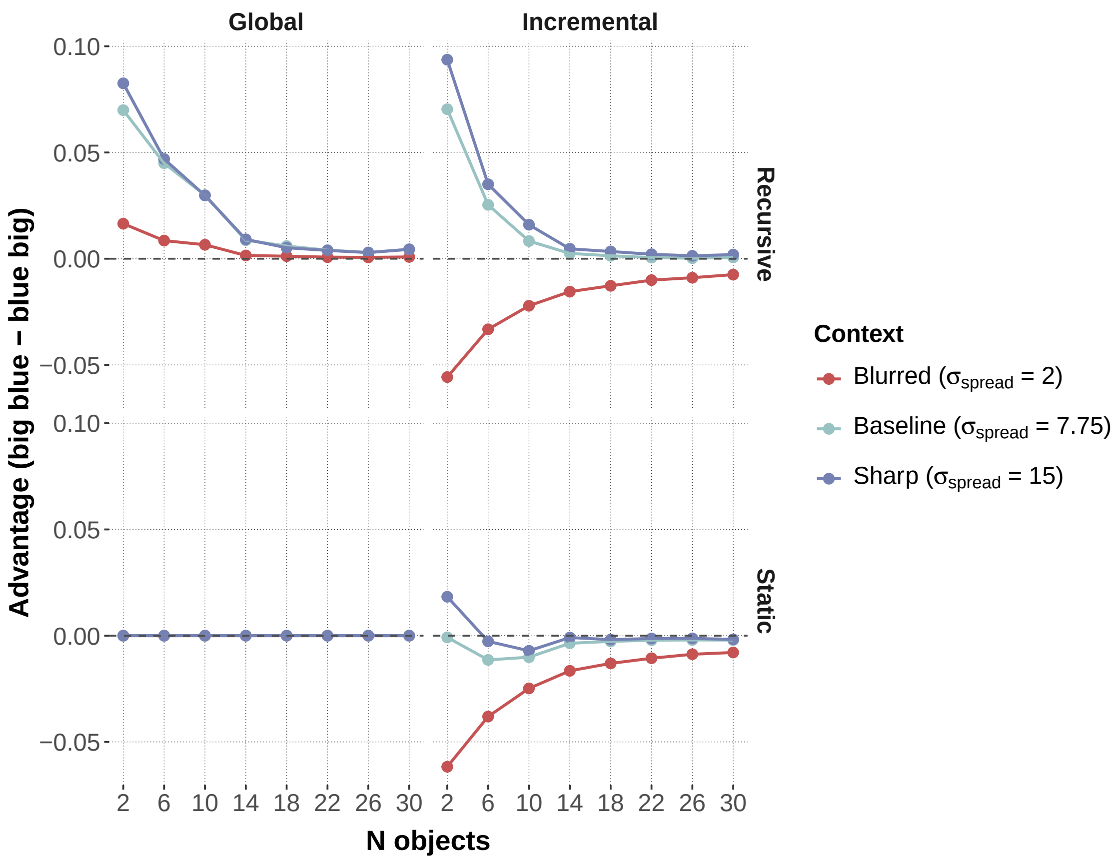

Figure 7

Size-first advantage as a function of context size (n_{obj}) and contextual spread (σ_{spread}), faceted by speaker model (columns) and semantic regime (rows). Under recursive semantics (top), both speakers favour size-first order, with the incremental speaker yielding a colour-first reversal in low-discriminability contexts. Under static semantics (bottom), the global speaker produces exactly zero advantage, while the incremental speaker defaults to colour-first.

generate a size-first advantage across most of the parameter space. A global speaker with context-sensitive composition already derives the canonical ordering asymmetry without any incremental machinery. This result is cumulative with the subjectivity literature: once semantic values are evaluated over progressively restricted comparison classes, less noisy inner modification is sufficient to produce the familiar ordering direction (Franke et al., 2019; Scontras et al., 2020, 2019).

Second, static semantics completely eliminates the global speaker's advantage. Under static semantics, the global speaker produced $\Delta = 0.000$ across all parameter combinations without exception. When the size threshold does not update during composition, the literal listener assigns identical posteriors to both orderings, making them informationally equivalent. This reveals that the global speaker's baseline advantage depends entirely on recursive semantics — it is not a property of the speaker architecture per se but of the order-sensitive threshold adaptation it exploits.

Third, the incremental speaker amplifies the size-first advantage under recursive semantics in much of the parameter space and, crucially, generates qualitatively distinct predictions in a specific region: low-discriminability, low-cardinality contexts, where colour-first preferences emerge. Under static semantics, the incremental speaker defaults to colour-first ($\Delta = -0.009$), driven by the chain-rule asymmetry that favours "blue" as the opening token when size is insufficiently diagnostic. This division of labour clarifies the architecture: context-sensitive composition provides the necessary asymmetry that the global speaker can exploit, while incremental production adds an online control policy whose ordering preference depends on the relative informativeness of each adjective at step 1.

7 General Discussion

The present results support a unified view of adjective ordering in referential communication. Three findings stand out. First, context-sensitive recursive composition is necessary for the baseline ordering effect: the simulation shows that the global speaker produces a size-first advantage only under recursive semantics, where threshold adaptation makes the two orderings informationally non-equivalent; under static semantics the advantage vanishes entirely. Second, incremental production is the dominant additional mechanism: both incremental models massively outperform both global models across both datasets. Third, context-updating semantics and incremental production interact — context-updating composition adds significant predictive value for production, but only when paired with an incremental speaker. The full model is not a stack of independent add-ons; its components are synergistic. These findings connect two strands of the ordering literature that have largely developed in parallel. The discriminatory-efficiency hypothesis (Fukumura, 2018) predicts that speakers place the most discriminating adjective first, and the empirical modulation of the size-first bias by referential context in both experiments is consistent with this prediction: size-first preferences increase when size is uniquely informative and weaken when colour carries more discriminatory load. The incremental-efficiency programme (Rubio-Fernández et al., 2021) goes further by proposing that ordering reflects an online production strategy rather than a stored preference, and the dominance of the incremental speaker models in both datasets supports precisely this claim. What the present model adds is an explicit mechanism linking the two accounts: incremental word-by-word decision-making is the production architecture that allows discriminatory strength to influence ordering in real time, and context-updating semantics is the compositional process that makes discriminatory strength order-dependent in the first place.

The interaction between semantic regime and speaker architecture is the most theoretically informative finding, and also the most nuanced. Context-updating semantics helps in the production data (15-way choice) but not in the slider data (binary choice), suggesting that the benefit depends on the richness of the alternative space. Within the global models, the complementary pattern holds in the slider data (context-updating helps) but not in production, where the LM prior already provides ordering information. This cross-pattern suggests that context-updating composition is most useful when the model lacks another source of order-relevant information and when the decision architecture can exploit non-commutative semantics step by step.

The production ablation models (reported in the Appendix) further illuminate the division of labour within the incremental speaker. The LM-only ablation, which strips out the RSA component, performs substantially worse than the full model, confirming that the corpus-derived prior alone cannot account for context-sensitive ordering. At the same time, the RSA-only ablation, which removes the LM prior entirely, failed PSIS-LOO diagnostics due to high Pareto-k values. This failure is itself informative: without a lexical prior to anchor the distribution over 15 utterance types, purely pragmatic choice produces degenerate predictive distributions that assign near-zero probability to many attested responses. The full model thus succeeds because the LM prior provides a well-shaped base distribution that the RSA mechanism reshapes in a context-sensitive direction — neither component is sufficient on its own. The lookahead ablation, which evaluates both remaining adjective orders before committing to each word, performed worse than the standard greedy strategy, suggesting that the myopic step-by-step policy is not only simpler but also a better characterisation of how speakers plan in this task.

Two limitations deserve mention. First, the computational analyses focus on the size–colour subset, although the experiments also included form-based trials. A fuller account would require either a stronger justification for treating form as interchangeable with colour or an expanded model that gives form a more explicit role. Second, the interaction between semantic regime and speaker architecture differs across tasks, which limits the generality of claims about when context-updating semantics is critical versus secondary. Tasks with richer alternative spaces may be needed to isolate its contribution more sharply. Third, the experiments tested only German-speaking participants and only size–colour adjective pairs. The model's semantic parameters (colour-similarity value, form-similarity value, word frequency) were fixed to corpus-derived estimates for German. Whether the same architecture generalises to languages with different canonical orderings (e.g., Romance postnominal adjectives; Cinque, 1994; Scott, 2002) or to adjective classes beyond the size–colour pair (e.g., age, material, shape) remains an open question. Cross-linguistic extension would require both new experimental data and language-specific semantic parameterisation.

Taken together, these results link adjective-ordering research to three traditions that are often discussed separately: subjectivity-based accounts of nominal modification, communicative-efficiency accounts of reference and redundancy, and incremental pragmatic models of language use. The contribution is not that one tradition replaces the others, but that they can be brought into a single probabilistic framework. Adjective ordering in referential communication is best understood as a graded adaptation to efficient, incremental reference resolution: subjectivity provides a stable prior over useful orders, discriminatory strength determines how the current scene reshapes that prior, and incremental production with context-updating semantics captures how speakers realise these pressures online.

This characterisation has implications for theories of nominal modification more broadly. Syntactic accounts that derive ordering from a fixed functional hierarchy (Cinque, 1994; Scott, 2002) correctly predict the default size-before-colour pattern but cannot explain the graded, context-dependent departures observed in both experiments. Subjectivity-based accounts (Franke et al., 2019; Scontras et al., 2017) provide a principled explanation for the default but treat it as a stable property of adjective classes rather than a context-sensitive outcome. The present framework subsumes both perspectives: the default ordering arises from the interaction between semantic noise and recursive composition (consistent with subjectivity accounts), while context-dependent modulation arises from the incremental speaker's sensitivity to the current discriminatory landscape (consistent with efficiency accounts). Future work should test whether this integration extends to adjective classes where subjectivity and efficiency predictions diverge — for instance, age or material adjectives in contexts where their discriminatory value can be experimentally manipulated.

8 References

Cinque, G. (1994). On the evidence for partial N-movement in the Romance DP. In G. Cinque, J. Koster, J.-Y. Pollock, L. Rizzi, & R. Zanuttini (Eds.), *Paths towards universal grammar: Studies in honor of Richard S. Kayne* (pp. 85–110). Washington, DC: Georgetown University Press.

CohnGordon, R., Goodman, N. D., & Potts, C. (2019). An incremental iterated response model of pragmatics. In *Proceedings of the society for computation in linguistics* (Vol. 2, pp. 81–90). doi: 10.7275/7ywv-jt08

Degen, J. (2023). The rational speech act framework. *Annual Review of Linguistics*, *9*, 519–540.

Degen, J., Hawkins, R. D., Graf, C., Kreiss, E., & Goodman, N. D. (2020). When redundancy is useful: A bayesian approach to "overinformative" referring expressions. *Psychological Review*, *127*(4), 591.

Dixon, R. M. W. (1982). *Where have all the adjectives gone? and other essays in semantics and syntax*. Berlin: Mouton.

Dyer, W., Torres, C., Scontras, G., & Futrell, R. (2023). Evaluating a century of progress on the cognitive science of adjective ordering. *Transactions of the Association for Computational Linguistics*, *11*, 1185–1200.

Feigenson, L., Dehaene, S., & Spelke, E. (2004). Core systems of number. *Trends in Cognitive Sciences*, *8*(7), 307–314.

Franke, M., Scontras, G., & Simonic, M. (2019). Subjectivity-based adjective ordering maximizes communicative success. In *Cogsci* (pp. 344–350).

Fukumura, K. (2018). Ordering adjectives in referential communication. *Journal of Memory and Language*, *101*, 37–50.

Futrell, R., Dyer, W., & Scontras, G. (2020, July). What determines the order of adjectives in English? comparing efficiency-based theories using dependency treebanks. In D. Jurafsky, J. Chai, N. Schluter, & J. Tetreault (Eds.), *Proceedings of the 58th annual* *meeting of the association for computational linguistics* (pp. 2003–2012). Online: Association for Computational Linguistics. Retrieved from `https://aclanthology.org/2020.acl-main.181/` doi: 10.18653/v1/2020.acl-main.181

Goodman, N. D., & Frank, M. C. (2016). Pragmatic language interpretation as probabilistic inference. *Trends in cognitive sciences*, *20*(11), 818–829.

Hahn, M., Degen, J., Goodman, N., Jurafsky, D., & Futrell, R. (2018). An information-theoretic explanation of adjective ordering preferences. In *Proceedings of the 40th annual meeting of the cognitive science society (cogsci)* (pp. 1766–1771). Retrieved from `https://cogsci.mindmodeling.org/2018/papers/0339/index.html`

Hoffman, M. D., & Gelman, A. (2014). The no-u-turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, *15*(47), 1593–1623.

Martin, J. E. (1969). Semantic determinants of preferred adjective order. *Journal of Verbal Learning and Verbal Behavior*, *8*(6), 697–704.

Phan, D., Pradhan, N., & Jankowiak, M. (2019). Composable effects for flexible and accelerated probabilistic programming in numpyro. *arXiv preprint arXiv:1912.11554*.

Quirk, R., Greenbaum, S., Leech, G., & Svartvik, J. (1972). *A grammar of contemporary english*. London: Longman.

Rubio-Fernández, P. (2016). How redundant are redundant color adjectives? an efficiency-based analysis of color overspecification. *Frontiers in psychology*, *7*, 153.

Rubio-Fernandez, P. (2019). Overinformative speakers are cooperative: Revisiting the gricean maxim of quantity. *Cognitive science*, *43*(11), e12797.

Rubio-Fernandez, P., & Jara-Ettinger, J. (2020). Incrementality and efficiency shape pragmatics across languages. *Proceedings of the National Academy of Sciences*, *117*(24), 13399–13404.

Rubio-Fernández, P., Mollica, F., & Jara-Ettinger, J. (2021). Speakers and listeners exploit word order for communicative efficiency: A cross-linguistic investigation. *Journal of Experimental Psychology: General*, *150*(3), 583–594.

Schlotterbeck, F., & Wang, H. (2023). An incremental rsa model for adjective ordering preferences in referential visual context. *Proceedings of the Society for Computation in Linguistics*, *6*(1), 121–132.

Schmidt, L. A., Goodman, N. D., Barner, D., & Tenenbaum, J. B. (2009). How tall is tall? Compositionality, statistics, and gradable adjectives. In N. A. Taatgen & H. van Rijn (Eds.), *Proceedings of the 31st annual conference of the cognitive science society.*

Scontras, G. (2023). Adjective ordering across languages. *Annual Review of Linguistics*, *9*, 357–376.

Scontras, G., Bar-Sever, G., Kachakeche, Z., Rosales Jr, C. M., & Samonte, S. (2020). Incremental semantic restriction and subjectivity-based adjective ordering. In *Proceedings of sinn und bedeutung* (Vol. 24, pp. 253–270).

Scontras, G., Degen, J., & Goodman, N. D. (2017). Subjectivity predicts adjective ordering preferences. *Open Mind*, *1*(1), 53–66.

Scontras, G., Degen, J., & Goodman, N. D. (2019). On the grammatical source of adjective ordering preferences. *Semantics and Pragmatics*, *12*, 7–1.

Scott, G.-J. (2002). Stacked adjectival modification and the structure of nominal phrases. In G. Cinque (Ed.), *Functional structure in DP and IP: The cartography of syntactic structures, vol. 1* (pp. 91–120). Oxford: Oxford University Press.

Sproat, R., & Shih, C. (1991). The cross-linguistic distribution of adjective ordering restrictions. In C. Georgopoulos & R. Ishihara (Eds.), *Interdisciplinary approaches to language: Essays in honor of S.-Y. Kuroda* (pp. 565–593). Kluwer.

Sweet, H. (1898). *A new english grammar, logical and historical, vol. 2: Syntax.* London: Clarendon.

Teodorescu, A. (2006). Adjective ordering restrictions revisited. In D. Baumer, D. Montero,

& M. Scanlon (Eds.), *Proceedings of the 25th West Coast Conference on Formal Linguistics* (pp. 399–407). Somerville, MA: Cascadilla Proceedings Project.

Trainin, N., & Shetreet, E. (2021). It's a dotted blue big star: on adjective ordering in a post-nominal language. *Language, Cognition and Neuroscience*, *36*(3), 320–341.

Trotzke, A., & Wittenberg, E. (2019). Long-standing issues in adjective order and corpus evidence for a multifactorial approach. *Linguistics*, *57*(2), 273–282. doi: 10.1515/ling-2019-0001

Vehtari, A., Gelman, A., & Gabry, J. (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, *27*(5), 1413–1432.

Waldon, B., & Degen, J. (2021). Modeling cross-linguistic production of referring expressions. In *Proceedings of the society for computation in linguistics 2021* (pp. 206–215). Association for Computational Linguistics.

Whorf, B. L. (1945). Grammatical categories. *Language*, *21*(1), 1–11.

Appendix A

Production Ablation Models

In addition to the four main model variants reported in the Model Comparison section, we fitted three ablation variants of the incremental speaker to the production data in order to isolate the contributions of specific architectural components. All ablation models use hierarchical participant pooling.

Incremental lookahead. Instead of choosing greedily at each word position, this variant evaluates each possible continuation by looking one step ahead and selecting the word that maximizes expected utility over the next two positions. It tests whether myopic word-by-word choice is sufficient or whether a short planning horizon improves fit.

LM only. This variant sets the RSA rationality parameter $\alpha = 0$, removing all pragmatic reasoning. Utterance choice is driven entirely by the language-model prior over word sequences, testing whether phrase-frequency information alone can account for ordering preferences.

RSA only. This variant sets the language-model weight $\beta = 0$, removing the utterance prior entirely. All ordering information must come from the RSA informativeness term and participant-level pragmatic strength. This model was successfully fit, but PSIS-LOO produced non-finite results because a substantial subset of observations have very high Pareto-k values ($k > 0.7$ for 199 of 3,196 observations). This indicates that the leave-one-out posterior is too different from the full posterior for reliable importance-sampling reweighting. The RSA-only model is therefore excluded from the ranked comparison but is noted here for completeness.

Figure A1 shows the ΔELPD comparison for the four main models and the two ablation variants with usable PSIS-LOO estimates. The incremental context-updating model remains clearly best. The lookahead variant performs worse than both global models, suggesting that greedy one-step choice is more appropriate than short-horizon planning for this task. The LM-only variant is the worst-fitting model overall, confirming that ordering preferences cannot be reduced to conventional phrase-frequency effects.

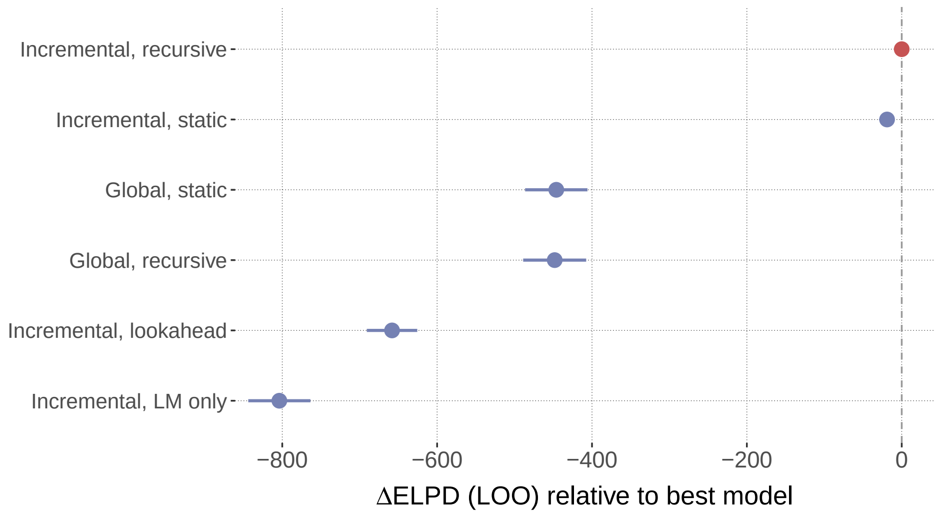

Figure A1

Production ablation model comparison. ΔELPD relative to the best model (incremental context-updating) with difference standard errors. The RSA-only ablation is excluded because PSIS-LOO produced non-finite estimates.

Appendix B

Simulation Parameter Sweeps

Figures B1 to B3 show how the three fixed semantic parameters affect communicative success across the two speaker models.

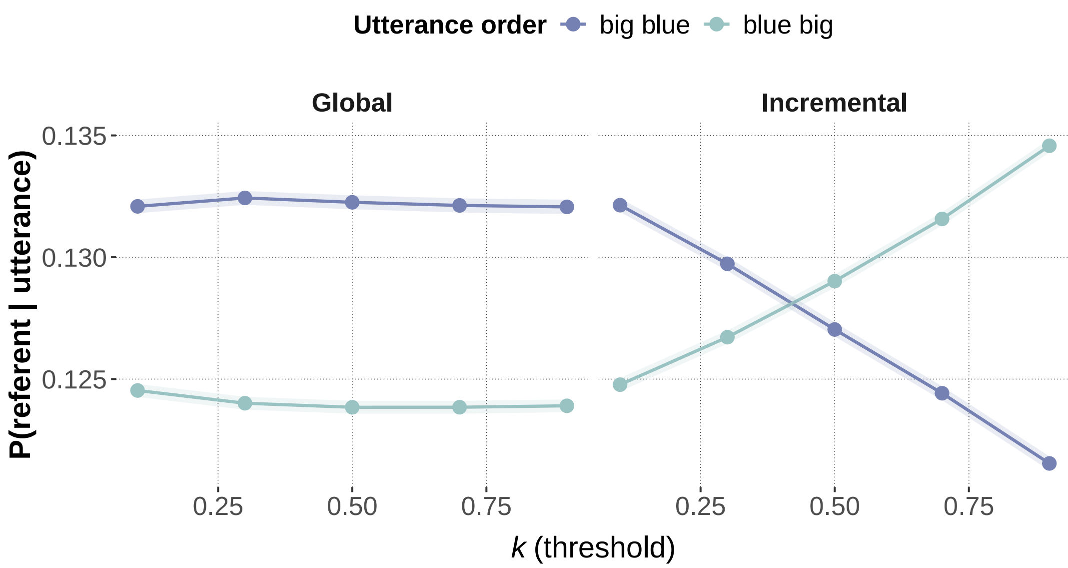

Figure B1

Effect of the threshold parameter k on communicative success in the simulation. Lower values of k make the size threshold more restrictive and reduce the success of both orderings, while preserving the incremental speaker's size-first advantage.

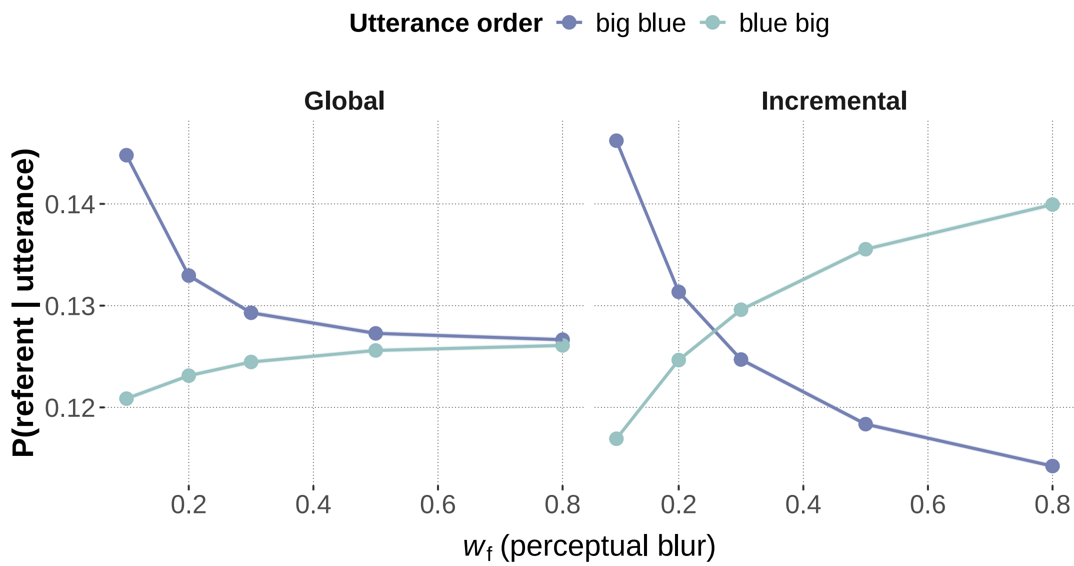

Figure B2

Effect of perceptual blur w_f on communicative success in the simulation. As blur increases, the advantage of size-first ordering decreases, especially for the global speaker.

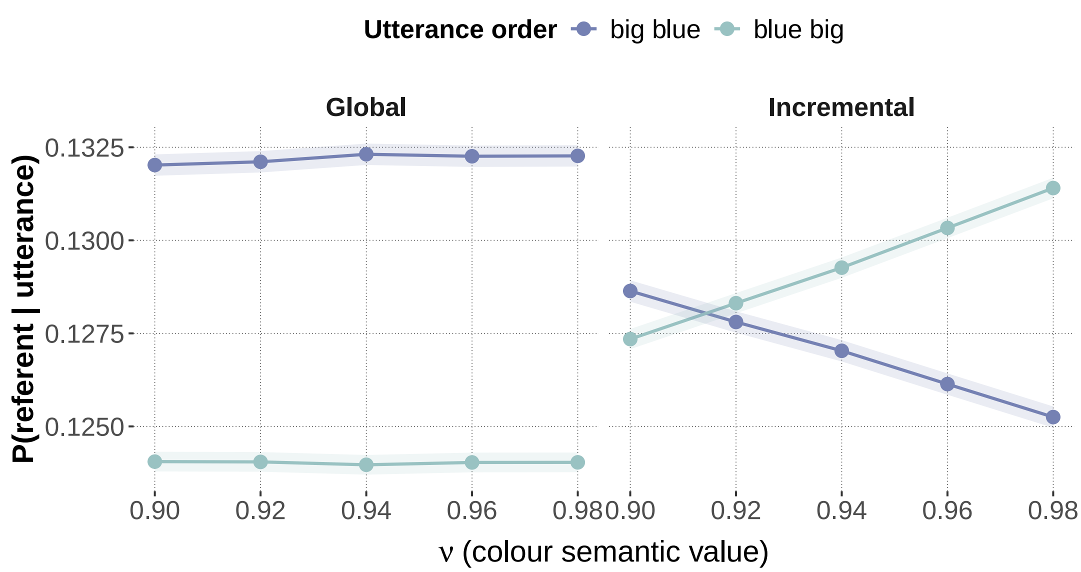

Figure B3

Effect of the colour/form semantic value ν on communicative success in the simulation. Higher values of ν make feature adjectives more deterministic and reduce the relative size-first advantage.

Appendix C

Posterior-Predictive Fit of the Best Model

Figures C1 and C2 show the condition-level posterior-predictive fit of the incremental context-updating model for both datasets.

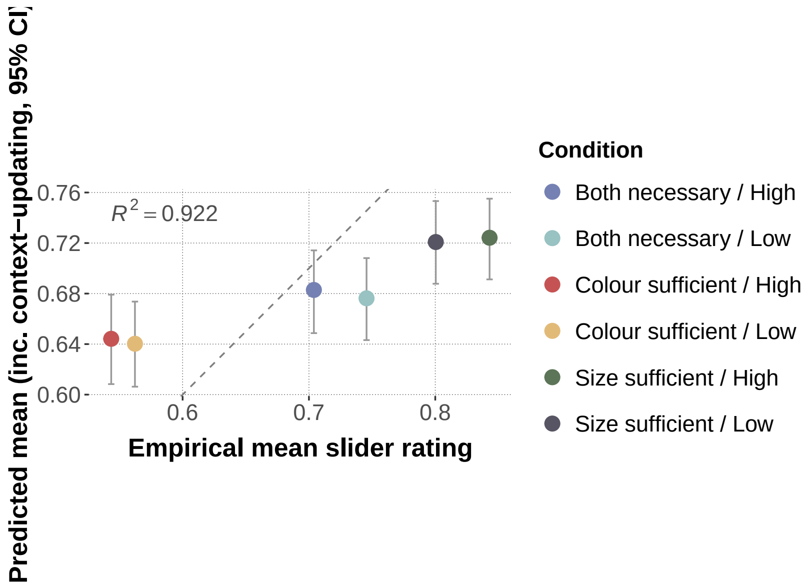

Figure C1

Slider-rating study: observed vs. predicted condition means for the incremental context-updating model. Error bars show 95% posterior-predictive intervals.

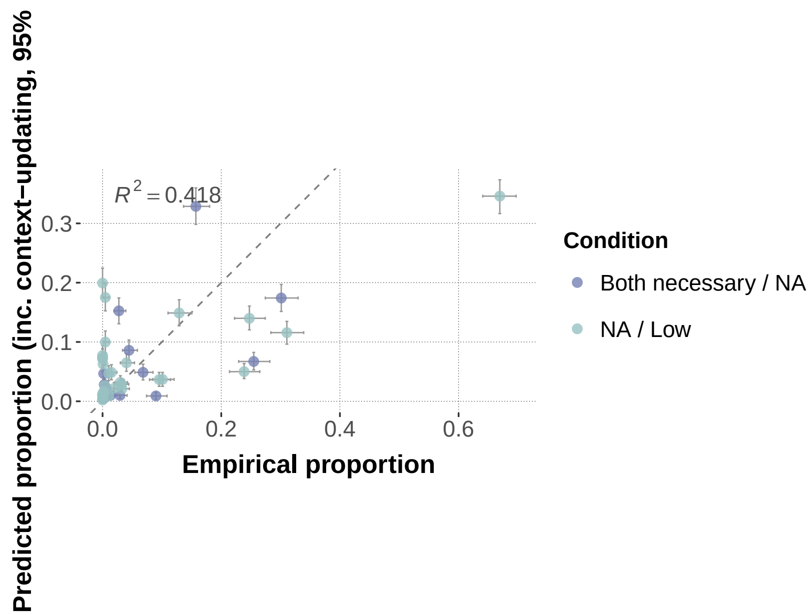

Figure C2

Free-production study: observed vs. predicted utterance-type proportions for the incremental context-updating model. Error bars show 95% posterior-predictive intervals.